Lab 7: Fuzzing for Input Validation Bugs

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Run the buggy application

Output of the original traffic_fine.py: valid input works; input "HELLO" crashes with a ValueError.

```
(venv) pv@pv-Satellite-S855:~/Documents/audit2$ python3 traffic_fine.py
==== Running parse_ticket_id ====
Enter a ticket ID (e.g. TF-2024-0456): TF-2024-0456
Ticket Prefix, Year and Serial: ('TF', 2024, 456)

==== Running total_fines ====
Enter fine amounts as CSV (e.g. 500,1200,300): 500, 1200, 300
Total Fines: 2000

==== Running speed_excess ====
Enter speed/limit (e.g. 72/60): 72/60
Percent Over Limit: 20
(venv) pv@pv-Satellite-S855:~/Documents/audit2$ python3 traffic_fine.py
==== Running parse_ticket_id ====
Enter a ticket ID (e.g. TF-2024-0456): HELLO
Traceback (most recent call last):
  File "/home/pv/Documents/audit2/traffic_fine.py", line 33, in <module>
    main()
  File "/home/pv/Documents/audit2/traffic_fine.py", line 19, in main
    result = parse_ticket_id(user_input)
             ^^^^^^^^^^^^^^^^^^^^^^^^^^^

  File "/home/pv/Documents/audit2/traffic_fine.py", line 2, in parse_ticket_id
    prefix, year, serial = ticket_id.split('-')
    ^^^^^^^^^^^^^^^^^^^^
ValueError: not enough values to unpack (expected 3, got 1)
```

Run test_traffic_fine.py against the buggy code

test_parse_ticket_id_no_crash fails: Hypothesis found 3 failures (bytes input, empty string, None).

```
(venv) pv@pv-Satellite-S855:~/Documents/audit2$ python -m pytest test_traffic_fine.py
============================================================ test session starts ============================================================
platform linux -- Python 3.12.3, pytest-9.2.0, pluggy-1.7.0
rootdir: /home/pv/Documents/audit2
plugins: hypothesis-6.170.15
collected 9 items

test_traffic_fine.py FFF..F.F.                                                                                                      [100%]

================================================================= FAILURES ==================================================================
_____________________________________________ test_parse_ticket_id_no_crash _____________________________________________
  + Exception Group Traceback (most recent call last):
  |   File "/home/pv/Documents/audit2/test_traffic_fine.py", line 10, in test_parse_ticket_id_no_crash
  |     def test_parse_ticket_id_no_crash(s):
  |                ^^^
  | ExceptionGroup: Hypothesis found 3 distinct failures. (3 sub-exceptions)
  +-+---------------- 1 ----------------
    | Traceback (most recent call last):
    |   File "/home/pv/Documents/audit2/test_traffic_fine.py", line 12, in test_parse_ticket_id_no_crash
    |     parse_ticket_id(s)
    |   File "/home/pv/Documents/audit2/traffic_fine.py", line 2, in parse_ticket_id
    |     prefix, year, serial = ticket_id.split('-')
    |                            ^^^^^^^^^^^^^^^^^^^^
    | TypeError: a bytes-like object is required, not 'str'
    |
    | During handling of the above exception, another exception occurred:
    |
    | Traceback (most recent call last):
    |   File "/home/pv/Documents/audit2/test_traffic_fine.py", line 14, in test_parse_ticket_id_no_crash
    |     assert False, f"parse_ticket_id crashed with: {e!r} on input {s!r}"
    | AssertionError: parse_ticket_id crashed with: TypeError("a bytes-like object is required, not 'str'") on input b''
    | assert False
    | Falling test case: test_parse_ticket_id_no_crash(
    |     s=b'',
    | )
```

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test_total_fines_no_crash fails: 3 failures (bytes input, empty string, None).

```
_____ test_total_fines_no_crash _____
  + Exception Group Traceback (most recent call last):
  |   File "/home/pv/Documents/audit2/test_traffic_fine.py", line 18, in test_total_fines_no_crash
  |     def test_total_fines_no_crash(s):
  |                ^^^
  | ExceptionGroup: Hypothesis found 3 distinct failures. (3 sub-exceptions)
  +-+---------------- 1 ----------------
    | Traceback (most recent call last):
    |   File "/home/pv/Documents/audit2/test_traffic_fine.py", line 20, in test_total_fines_no_crash
    |     total_fines(s)
    |   File "/home/pv/Documents/audit2/traffic_fine.py", line 6, in total_fines
    |     amounts = [int(a) for a in amounts_csv.split(',')]
    |                               ^^^^^^^^^^^^^^^^^^^^^^^
    | TypeError: a bytes-like object is required, not 'str'
    |
    | During handling of the above exception, another exception occurred:
    |
    | Traceback (most recent call last):
    |   File "/home/pv/Documents/audit2/test_traffic_fine.py", line 22, in test_total_fines_no_crash
    |     assert False, f"total_fines crashed with: {e!r} on input {s!r}"
    | AssertionError: total_fines crashed with: TypeError("a bytes-like object is required, not 'str'") on input b''
    | assert False
    | Falling test case: test_total_fines_no_crash(
    |     s=b'',
    | )
```

test_speed_excess_no_crash fails: 4 failures (bytes input, "/", empty string, None).

```
_____ test_speed_excess_no_crash _____
  + Exception Group Traceback (most recent call last):
  |   File "/home/pv/Documents/audit2/test_traffic_fine.py", line 26, in test_speed_excess_no_crash
  |     def test_speed_excess_no_crash(s):
  |                ^^^
  | ExceptionGroup: Hypothesis found 4 distinct failures. (4 sub-exceptions)
  +-+---------------- 1 ----------------
    | Traceback (most recent call last):
    |   File "/home/pv/Documents/audit2/test_traffic_fine.py", line 28, in test_speed_excess_no_crash
    |     speed_excess(s)
    |   File "/home/pv/Documents/audit2/traffic_fine.py", line 13, in speed_excess
    |     speed, limit = reading.split('/')
    |                    ^^^^^^^^^^^^^^^^^^
    | TypeError: a bytes-like object is required, not 'str'
    |
    | During handling of the above exception, another exception occurred:
    |
    | Traceback (most recent call last):
    |   File "/home/pv/Documents/audit2/test_traffic_fine.py", line 30, in test_speed_excess_no_crash
    |     assert False, f"speed_excess crashed with: {e!r} on input {s!r}"
    | AssertionError: speed_excess crashed with: TypeError("a bytes-like object is required, not 'str'") on input b''
    | assert False
    | Falling test case: test_speed_excess_no_crash(
    |     s=b'',
    | )
```

Summary: 5 failed, 4 passed.

```
=========== short test summary info ===========
FAILED test_traffic_fine.py::test_parse_ticket_id_no_crash - ExceptionGroup: Hypothesis found 3 distinct failures. (3 sub-exceptions)
FAILED test_traffic_fine.py::test_total_fines_no_crash - ExceptionGroup: Hypothesis found 3 distinct failures. (3 sub-exceptions)
FAILED test_traffic_fine.py::test_speed_excess_no_crash - ExceptionGroup: Hypothesis found 4 distinct failures. (4 sub-exceptions)
FAILED test_traffic_fine.py::test_total_fines_result_never_exceeds_cap - ValueError: invalid literal for int() with base 10: ''
FAILED test_traffic_fine.py::test_known_bad_inputs_return_none - AttributeError: 'NoneType' object has no attribute 'split'
=========== 5 failed, 4 passed in 4.73s ===========
```

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Run the fixed application

Fixed traffic_fine.py: valid input still works; invalid input ("HELLO") is handled gracefully.

```
(venv) pv@pv-Satellite-S855:~/Documents/audit2$ python3 traffic_fine.py
==== Running parse_ticket_id ====
Enter a ticket ID (e.g. TF-2024-0456): TF-2024-0456
Ticket Prefix, Year and Serial: ('TF', 2024, 456)

==== Running total_fines ====
Enter fine amounts as CSV (e.g. 500,1200,300): 50,1200,300
Total Fines: 1550

==== Running speed_excess ====
Enter speed/limit (e.g. 72/60): 72/60
Percent Over Limit: 20
(venv) pv@pv-Satellite-S855:~/Documents/audit2$ python3 traffic_fine.py
==== Running parse_ticket_id ====
Enter a ticket ID (e.g. TF-2024-0456): HELLO
Ticket Prefix, Year and Serial: Invalid input

==== Running total_fines ====
Enter fine amounts as CSV (e.g. 500,1200,300):
```

Re-run test_traffic_fine.py against the fixed code

All 9 tests pass.

```
(venv) pv@pv-Satellite-S855:~/Documents/audit2$ python -m pytest test_traffic_fine.py -v
============================= test session starts ==============================
platform linux -- Python 3.12.3, pytest-9.2.0, pluggy-1.7.0 -- /home/pv/Documents/audit2/venv/bin/python
cachedir: .pytest_cache
hypothesis profile 'default'
rootdir: /home/pv/Documents/audit2
plugins: hypothesis-6.170.15
collected 9 items

test_traffic_fine.py::test_parse_ticket_id_no_crash PASSED               [ 11%]
test_traffic_fine.py::test_total_fines_no_crash PASSED                   [ 22%]
test_traffic_fine.py::test_speed_excess_no_crash PASSED                  [ 33%]
test_traffic_fine.py::test_parse_ticket_id_valid_roundtrip PASSED        [ 44%]
test_traffic_fine.py::test_total_fines_valid_is_capped_sum PASSED        [ 55%]
test_traffic_fine.py::test_total_fines_result_never_exceeds_cap PASSED   [ 66%]
test_traffic_fine.py::test_speed_excess_valid PASSED                     [ 77%]
test_traffic_fine.py::test_known_bad_inputs_return_none PASSED           [ 88%]
test_traffic_fine.py::test_known_good_inputs PASSED                      [100%]

============================== 9 passed in 0.81s ===============================
```